Finite-word contraction and short descent certificates for the Juggler map

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Abstract

The Juggler map is the nonlinear integer map

$$J(n) = \begin{cases} \lfloor \sqrt{n} \rfloor, & n \text{ even}, \\ \lfloor n^{3/2} \rfloor, & n \text{ odd}. \end{cases}$$

It is conjectured that every positive integer eventually reaches 1. We do not prove that conjecture.

For every finite parity word w realized at n, the iterate satisfies the power envelope

$$J^{|w|}(n)^{2^{|w|}} \le n^{3^{\#O(w)}}.$$

Hence $3^{\#O(w)} < 2^{|w|}$ forces $J^{|w|}(n) < n$ whenever $n \geq 2$. An exact global-defect identity records the floor slack in this comparison. These statements are conditional on a realized word: they do not say that every orbit meets a contracting word.

A start $n \geq 2$ has a *descent certificate* if some realized finite word sends it strictly below n. Every even start, and every odd start whose first image is even, has a uniform one- or two-step certificate. A classical discrepancy bound

$$\left|S_O(N)\right| \ll N^{5/6}, \qquad S_O(N) = \sum_{\substack{n \leq N \\ n \text{ odd}}} (-1)^{\lfloor n^{3/2} \rfloor},$$

implies that the complementary odd-to-odd starts have density 1/4, so the uniform short-certificate class has density 3/4. This is not a density of all descent certificates, and it is not a density of starts that reach 1. Odd-to-odd starts expand on the first step and may still descend later. The bound itself is ambient: it does not transfer to sparse Juggler-generated image sets.

The remaining question, analogue of Terras's almost-all stopping-time theorem, is whether almost every odd-to-odd start has some finite descent. No such theorem is proved here.

1. Introduction

The Juggler sequence was introduced by Pickover [1]; see also Weisstein [2] and OEIS A007320 [3]. Universal arrival at 1 remains open. In this repository every start $n \leq 4000$ reaches 1, as an exact finite computation. That check is not a proof.

The map combines a contracting even branch with an expanding odd branch,

$$E(n) = \lfloor n^{1/2} \rfloor, \qquad O(n) = \lfloor n^{3/2} \rfloor.$$

A word of length k with o odd letters has ideal exponent $3^o/2^k$. Floors are applied after every letter, and a word is available only when the orbit realizes those parities. The paper records the exact finite-word comparison, the exact slack, the uniform short certificates, and the density of the class those certificates cover.

The contribution is a short exact-arithmetic note, not a survey of failed compressions and not a description of the computational atlas used in the surrounding laboratory. We do not prove a Collatz theorem, and we do not transfer Collatz stopping-time results to J.

1.1 Evidence discipline

- **EXACT — LEAN VERIFIED:** a theorem checked by Lean in the software archive named in Section 7.
- **EXACT — HUMAN PROOF:** a mathematical argument not packaged in Lean. Theorem 5.1 is of this kind.
- **COMPUTATIONALLY VERIFIED:** an exact finite computation under stated bounds.
- **OBSERVATION:** a descriptive pattern in finite data.
- **REFUTED:** a universal candidate killed by a certificate.

A finite check is never called a termination proof.

1.2 Related maps

The nearest published comparison is the Collatz problem. Lagarias [4,5] surveys parity words, stopping times, and almost-all statements that stop short of totality. Terras [6] and Everett [7] proved that almost every positive integer has a finite Collatz stopping time — some return below the start — without proving that every orbit reaches 1. Tao [8] later showed that almost all Collatz orbits attain almost bounded values.

Those results are cited as methodological cousins, not as theorems about J. The uniform short certificates below cover a set of density 3/4 by a one- or two-step argument. Terras and Everett prove that almost every Collatz start has *some* finite stopping time. Those are different statements. The Juggler analogue of Terras would be an almost-all descent theorem on the odd-to-odd class. It is not proved here.

Crandall [9] and Matthews–Watts [10] treat piecewise-affine Hasse–Syracuse maps. Juggler is not piecewise affine: the branches are floor powers. Prasad and Prasad [12] estimate excursion and stopping constants for juggler-like random walks; Section 6 keeps that comparison descriptive.

2. Finite words, envelope, and defect

Let $\mathcal{B} = \{E, O\}$. A finite word $w \in \mathcal{B}^*$ is *realized* at $n \in \mathbb{N}$ when the successive parities of the orbit of n are exactly the letters of w. Write $J^{|w|}(n)$ for the endpoint after those letters, and $\#O(w)$ for the number of odd letters.

The identities in this section are formalized in Lean (`follows_iff_word`, `image_eq_iterate`, `image_append`, `image_monotone_of_follows`, `power_bound_word`, `power_bound_contracts`, `global_defect_identity`). The proofs below are the ordinary integer arguments.

Theorem 2.1 (fixed-word monotonicity; EXACT — LEAN VERIFIED). If $n \le m$ and both realize w, then $J^{|w|}(n) \le J^{|w|}(m)$.

The realizing set of a fixed word need not be an interval.

Theorem 2.2 (finite-word power envelope; EXACT — LEAN VERIFIED). If w is realized at n and $m = J^{|w|}(n)$, then

$$m^{2^{|w|}} \le n^{3^{\#O(w)}}.$$

Proof. Write $k = |w|$ and $o = \#O(w)$. The empty word is the equality $n \le n$. Suppose the bound holds at a realized prefix ending at x, and the next letter is realized.

If the next letter is even, then $J(x) = \lfloor \sqrt{x} \rfloor$, so $J(x)^2 \le x$. Raising the inductive bound to the second power gives

$$J(x)^{2^{k+1}} = (J(x)^2)^{2^k} \le x^{2^k} \le n^{3^o}.$$

The odd count is unchanged.

If the next letter is odd, then $J(x) = \lfloor x^{3/2} \rfloor$, so $J(x)^2 \le x^3$. Therefore

$$J(x)^{2^{k+1}} = (J(x)^2)^{2^k} \le x^{3 \cdot 2^k} = (x^{2^k})^3 \le \left(n^{3^o}\right)^3 = n^{3^{o+1}}.$$

This is the claimed bound after one more odd letter. □

Corollary 2.3 (exponent-gap contraction; EXACT — LEAN VERIFIED). If $n \ge 2$, w is realized at n, and $3^{\#O(w)} < 2^{|w|}$, then $J^{|w|}(n) < n$.

Proof. Let $m = J^{|w|}(n)$ and $k = |w|$. Theorem 2.2 gives $m^{2^k} \le n^{3^{\#O(w)}}$. The exponent gap and $n \ge 2$ give $n^{3^{\#O(w)}} < n^{2^k}$, so $m^{2^k} < n^{2^k}$. Since $m \ge 1$, one has $m < n$. □

The corollary includes familiar contracting blocks such as *OOOEE*. It does not prove that every start realizes some contracting word.

The floor slack is exact. For a single branch,

$$x^e = J(x)^2 + \rho(x), \qquad e = \begin{cases} 1, & x \text{ even}, \\ 3, & x \text{ odd}, \end{cases}$$

with $0 \le \rho(x) < 2J(x) + 1$. Lifting these remainders through a word gives a nonnegative global defect $\Delta_w(n)$.

Theorem 2.4 (global defect identity; EXACT — LEAN VERIFIED). If w is realized at n and $m = J^{|w|}(n)$, then

$$n^{3^{\#O(w)}} = m^{2^{|w|}} + \Delta_w(n), \qquad \Delta_w(n) \ge 0.$$

The identity names the slack in Theorem 2.2. It does not supply a state-independent positive tax: persistent expanding blocks can have arbitrarily small observed normalized slack at large scale.

3. Inverse cells and cycles

The exact one-step fibers are

$$J(n) = q \iff q^2 \le n < (q+1)^2 \quad (n \text{ even})$$

and

$$J(n) = m \iff m^2 \le n^3 < (m+1)^2 \quad (n \text{ odd}).$$

An odd fiber contains at most one integer (`odd_cell_unique`; EXACT — LEAN VERIFIED). An even fiber is a parity-restricted square interval and may contain many predecessors.

A nonempty realized word w with $J^{|w|}(n) = n$ is a cycle word.

The identities in this section are formalized in Lean (`cycle_word_formally_expanding`, `cycleMin_not_end_odd`, `no_cycle_word_oooeoe`, `no_cycle_word_ooooee`). The proofs below are the ordinary integer arguments.

Theorem 3.1 (cycle exponent condition; EXACT — LEAN VERIFIED). Every cycle word at $n \ge 2$ satisfies

$$2^{|w|} < 3^{\#O(w)}.$$

Thus a contracting word cannot close a nontrivial cycle. The cycle minimum is odd and the cycle maximum is even; a minimum-based orientation cannot end in an odd letter. These are necessary restrictions. They do not exclude every nontrivial cycle.

The two remaining legal minimum-based length-six orientations are nevertheless impossible. The argument uses the last-even cell together with a coarse lower envelope. For $n \ge 1$,

$$n < 4\,\lfloor\sqrt{n}\rfloor^2,$$

so an even step satisfies $n \le 4\,J(n)^2$ and an odd step satisfies $n^3 \le 4\,J(n)^2$. Composing along a realized word v of length k with odd count o gives

$$n^{3^o} \le D_v\, J^k(n)^{2^k}.$$

The denominator starts at 1 and updates by $D \mapsto D \cdot 4^{2^j}$ on an even letter and $D \mapsto D^3 \cdot 4^{2^j}$ on an odd letter, at step j. In particular $D_{OOO} = 2^{38}$ and $D_{OOOO} = 2^{130}$. If a cycle word ends in an even letter, the last-even cell is $J^{|w|-1}(n) < (n+1)^2$.

Theorem 3.2 (leftover length-six orientations; EXACT — LEAN VERIFIED). Neither *OOOEOE* nor *OOOOEE* is a cycle word at any $n \ge 2$. This is not an exclusion of every length-six word, nor of every cycle.

Proof. First, if $n \ge 256$, then

$$n^{81} > 2^{130}(n+1)^{64}.$$

Indeed $257^{64} < 2 \cdot 256^{64}$ and $256(n+1) \le 257\,n$, so $(n+1)^{64} < 2n^{64}$. Then $2^{130}(n+1)^{64} < 2^{131}n^{64}$. Since $n \ge 256 = 2^8$, one has $n^{17} \ge 2^{136}$, and therefore $2^{131}n^{64} < 2^{136}n^{64} \le n^{81}$.

For $n < 256$, neither word returns to its start: this is an exhaustive evaluation of the two itineraries.

Now suppose $n \ge 256$ realizes *OOOOEE*, and write $z = J^4(n)$ for the image after the prefix *OOOO*. The last two even letters give $J(z) < (n+1)^2$ and $z < (J(z)+1)^2$, hence $z < (n+1)^4$. The lower envelope on *OOOO* is $n^{81} \le 2^{130}z^{16}$, so $n^{81} < 2^{130}(n+1)^{64}$, contradicting the tail inequality.

Finally suppose $n \ge 256$ realizes *OOOEOE*. Write $z_3 = J^3(n)$ and $y = J(z_3) = \lfloor\sqrt{z_3}\rfloor$, so $z_3 < (y+1)^2$. The lower envelope on *OOO* is $n^{27} \le 2^{38}z_3^8 < 2^{38}(y+1)^{16}$. Cubing yields $n^{81} < 2^{114}(y+1)^{48}$. The last letters *OE* give the odd-cell bound $y^3 < (n+1)^4$. Write $A = n+1 \ge 257$. Then $(y+1)^3 < 2A^4$: if $y \le A$, this follows from $(A+1)^3 < 2A^4$; if $y > A$, then $y \ge 4$ and $(y+1)^3 = y^3 + 3y^2 + 3y + 1 < A^4 + 4y^2$, while $4y^2 < A^4$ because $4y^3 < 4A^4 \le yA^4$. Raising $(y+1)^3 < 2A^4$ to the sixteenth power gives $(y+1)^{48} < 2^{16}(n+1)^{64}$. Combining with the cubed lower envelope produces again $n^{81} < 2^{130}(n+1)^{64}$. □

4. Short descent certificates

A start $n \ge 2$ has a *descent certificate* if there exists a realized finite word w with $J^{|w|}(n) < n$, or with image 1. In the Lean development this predicate is called `FiniteProgress`. It is existential over all finite words, not only over words of length one or two.

Theorem 4.1 (uniform short certificates; EXACT — LEAN VERIFIED). Let $n \ge 2$.

1. If n is even, then the one-letter word E is a descent certificate: $J(n) = \lfloor\sqrt{n}\rfloor < n$.
2. If n is odd and $J(n)$ is even, then the two-letter word OE is a descent certificate.

Theorem 4.2 (unresolved starts are odd-to-odd; EXACT — LEAN VERIFIED). If $n \ge 2$ has no descent certificate, then n is odd and $J(n)$ is odd.

The converse is false. Theorem 4.2 does not say that odd-to-odd starts lack descent certificates. Many of them descend after a longer word.

If every start above 1 has some descent certificate, ordinary strong induction yields arrival at 1 (`reachesOne_of_all_finiteProgress`). The hypothesis is not proved. A certificate at n only reduces the problem to a strictly smaller positive integer, which may itself be odd-to-odd.

Lean also certifies a finite landing class (`reachesOne_of_lt_twelve`, `even_lt_sq_twelve_reachesOne`):

Theorem 4.3 (small residuals; EXACT — LEAN VERIFIED). Every $y \in \{1, \dots, 11\}$ reaches 1. Consequently every even residual strictly below $144 = 12^2$ reaches 1.

This enlarges the set of fatal landings for a hypothetical minimal counterexample. It does not cover all odd-to-odd starts, and it does not prove that every even start reaches 1.

Independently, an exact computation gives:

Proposition 4.4 (window totality; COMPUTATIONALLY VERIFIED). Every start $1 \leq n \leq 4000$ reaches 1.

This is a finite check, not a bound on all starts.

On the complementary odd-to-odd class, first return below the start is frequent at a short horizon, but not automatic.

Proposition 4.5 (odd-to-odd first return; OBSERVATION). For starts $2 \leq n \leq N$ and horizon 20, write OO for the odd-to-odd starts in that range. The exact census is:

N	#OO	OO first return ≤ 20	all starts, first return ≤ 20
10^3	252	0.877	0.969
10^4	2504	0.887	0.972
10^5	24984	0.896	0.974
10^6	249926	0.895	0.974

At $N = 10^6$, 26,243 odd-to-odd starts have no return below the start in 20 steps. None of these rows is a Lean descent certificate, and none is an almost-all theorem.

5. Ambient discrepancy and the short-certificate class

For odd n write $s(n) = (-1)^{\lfloor n^{3/2} \rfloor}$ and

$$S_O(N) = \sum_{\substack{n \leq N \\ n \text{ odd}}} s(n).$$

Let $M(N)$ be the number of odd integers $n \leq N$, and let

$$\mathrm{OO}(N) = \#\{n \leq N : \ n \text{ odd}, \ J(n) \text{ odd}\}.$$

Then $s(n) = -1$ if and only if $J(n)$ is odd, and

$$S_O(N) = M(N) - 2\,\mathrm{OO}(N).$$

Also $M(N) = N/2 + O(1)$.

The floor sign has an exact fractional-part form. Write $x = \lfloor x \rfloor + \{x\}$. If $\lfloor x \rfloor$ is even then $\{x/2\} < 1/2$; if $\lfloor x \rfloor$ is odd then $\{x/2\} \geq 1/2$. Thus

$$\lfloor x \rfloor \text{ is odd} \quad \Longleftrightarrow \quad \{x/2\} \geq \tfrac{1}{2}.$$

For odd $n = 2r + 1$ set
$$g(r) = \frac{(2r+1)^{3/2}}{2}.$$
Then $s(n) = -1$ if and only if $\{g(r)\} \geq 1/2$, and $S_O(N)$ is twice the interval discrepancy of the sequence $\{g(r)\}$ against $[1/2, 1)$.

The following two lemmas are classical; see Kuipers–Niederreiter [11, Ch. 1–2].

Lemma 5.A (van der Corput, second-derivative form). Let f be twice differentiable on an interval of length M, with $\lambda \leq |f''| \leq \alpha\lambda$ for some $\alpha \geq 1$. Then
$$\Big|\sum e(f)\Big| \ll_\alpha M\lambda^{1/2} + \lambda^{-1/2},$$
where the sum runs over the integers in the interval and $e(t) = e^{2\pi i t}$.

Lemma 5.B (Erdős–Turán). The discrepancy of R points $x_j \in \mathbb{R}/\mathbb{Z}$ against an interval satisfies
$$D \ll \frac{R}{H} + \sum_{h=1}^{H} \frac{1}{h}\Big|\sum_{j=1}^{R} e(hx_j)\Big|$$
for every cutoff $H \geq 1$.

Theorem 5.1 (ambient odd-input discrepancy; EXACT — HUMAN PROOF).
$$|S_O(N)| \ll N^{5/6}.$$
This argument is not packaged in Lean.

Proof. The second derivative is
$$g''(r) = \tfrac{3}{2}(2r+1)^{-1/2}.$$
It is positive and decreasing. On a dyadic block $r \asymp M$ one has $g''(r) \asymp M^{-1/2}$. For the h-th Fourier mode take $f = hg$, so $\lambda \asymp hM^{-1/2}$. Lemma 5.A gives
$$\Big|\sum_{r \asymp M} e(hg(r))\Big| \ll h^{1/2}M^{3/4} + h^{-1/2}M^{1/4}.$$
Lemma 5.B, with $R \asymp M$ and the relation between S_O and interval discrepancy, yields
$$|S_O| \ll \frac{M}{H} + M^{3/4}H^{1/2} + M^{1/4}$$
on that block: the middle term is $\sum_{h \leq H} h^{-1} \cdot h^{1/2}M^{3/4} \ll M^{3/4}H^{1/2}$, and the last term is $\sum_{h \leq H} h^{-1} \cdot h^{-1/2}M^{1/4} \ll M^{1/4}$. The choice $H \asymp M^{1/6}$ balances the first two terms at $M^{5/6}$. Summing dyadic blocks up to N preserves the exponent. The floor sign is not replaced by a single exponential; the exponential sums are those of the sequence $\{g(r)\}$. □

Corollary 5.2 (short-certificate class; EXACT — HUMAN PROOF). The odd-to-odd starts have natural density 1/4:
$$\big|\mathrm{OO}(N) - N/4\big| \ll N^{5/6}.$$
Equivalently, the starts that admit the uniform one- or two-step certificates of Theorem 4.1 — every even start together with every odd-to-even start — have natural density 3/4.

This is a counting corollary of Theorem 5.1, not a Lean cardinality theorem. It is a density of a *uniform short-certificate class*. It is not the density of all starts that possess some descent certificate, and it is not a density of starts that reach 1.

The exact census through $N = 10^7$ has $\mathrm{OO}(N)/N = 0.2499896$. Through $N = 10^6$ one has $S_O(N) = 146$ and running maximum 256, at $n = 985351$. A spot computation at 10^7 has running maximum 459. The observed growth is much smaller than $N^{5/6}$. The descriptive $N^{1/3}$-scale envelope is not promoted to a theorem.

The bound does not transfer from intervals to Juggler images. For a general interval $I = [A, B]$ the variation of S_O depends on the right endpoint, not only on $|I|$. The expanding image $Y = J_O(O(I))$ is a sparse gap set. At source bound 10^6, recorded consecutive gaps in Y range from 4 to 3000. A certified finite concentration is the run of 52 consecutive odd sources of equal sign on $[952525, 952627]$. Therefore ambient interval cancellation does not imply orbit or image-set cancellation. This is a boundary for the present argument, not a proof that no sparse-sequence estimate can exist.

6. The remaining gap

Theorem 4.1 and Corollary 5.2 together say that a uniform one- or two-step argument covers a set of density 3/4. The first odd-to-odd image expands, so ordinary strong induction cannot fire on that class. Proposition 4.5 shows that most odd-to-odd starts in a finite window still return below the start inside twenty steps. That is an observation, not Terras's theorem for J.

A mixed-parity heuristic, ignoring floors, gives mean log-log drift $\frac{1}{2}\log(3/4) < 0$. Finite ensembles sit near this value; hard paths are more odd-rich. This agrees qualitatively with the juggler-like random-walk model of Prasad and Prasad [12]. Fair parity is an assumption, not a dynamical theorem, and typical negative drift is not pointwise contraction.

The gap is therefore:

> No theorem forces every exact integer state into a contracting prefix. In particular, it is open whether almost every odd-to-odd start has a finite descent certificate.

That is the Juggler form of the Terras question. It is the next mathematical target. It is not answered by enlarging the finite-word atlas, by another residual quotient, or by a sharper ambient discrepancy exponent.

7. Software archive

Lean proofs of the exact-arithmetic theorems, the computational checks, and the laboratory notes live at https://github.com/sneakyweasel/balanced_ternary/. The archive is not required to read the arguments above. The discrepancy bound is a human proof; Lean does not certify it.

From a clone, the formal layer and the focused records are

```
pip install -e ".[dev]"
cd formal && lake build
python tools/render_theorem_ledger.py --check
python -m pytest tests/unit/test_theorem_ledger.py
python -m pytest tests/research/juggler_sequence/test_progress_coverage.py
```

```
python -m pytest tests/research/juggler_sequence/test_odd_image_discrepancy.py
python -m pytest tests/research/juggler_sequence/test_global_defect.py
python -m pytest tests/research/juggler_sequence/test_cycle_leftover_words.py
```

A Word Atlas used in the laboratory records bounded realizers and persistent-expanding blocks through length 20 and starts $n \leq 10^8$. It is infrastructure, not a theorem of this note. Absence of a word in that census means only that the word was not observed inside the search bound.

Acknowledgments

I used large language models extensively while drafting and revising the text, organizing companion notes, and as an interactive assistant for Lean statements, tests, and literature records. The models are not authors. Lean theorems and named computations are the certificates for those claims. The discrepancy bound (Theorem 5.1) is a human proof using classical analytic inequalities; it is not Lean-certified. I take full responsibility for the contents.

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